

Assignment 1

Rules:

- You may use L^AT_EX, Word, or any other text editor but please hand in a **pdf document** with your solution.
- It is recommended to use Python or Ox for solving the assignment. Alternatively, you may use **R** or Matlab. You cannot base your report on output of Eviews, Stata, or such.
- You must write your own code - built-in packages (such as **vars** in R) may only be used to double-check the results. **Please hand in your code in an Appendix. Make sure it is well-documented.**
- You should **answer** the questions - do not just paste estimation output without explanation (except when nothing else is asked). **All results should be presented in a readable, clear and concise way.**
- You should hand in one pdf document for each group of **4 students**.
- The deadline is Friday 27 February at 23:59 hours.

In this assignment, you are asked to carry out a panel data analysis for the “Cornwell and Rupert” data set on “Returns to Schooling Data”. The main research question is whether more schooling leads to higher salaries later on. Download the data set **data_assignment1** from Canvas and load it into your software. There is an accompanying file “ReadmeCornwellRupertData.txt” with further information about the variables in the data file and how the data is organized. The data set includes seven years (1976-1982) of 595 individuals ($T = 7$, $N = 595$, $N \times T = 4165$) with variable log-wages and a set of explanatory variables.

Pre-assignment work:

- Carry out a simple regression for each year for the data of the 595 individuals (i):

$$\log \text{ wages}_i = \text{constant} + \text{all variables } X_i' \beta + \varepsilon_i.$$

Report the least squares estimates of constant and β in a table with seven columns for the years 1976-1982, include standard errors and/or t-statistics. Comment on your findings.

- Carry out a pooled regression (same model) for all years, that is for all data of 4165 observations. Comment on your findings and relate these to your yearly results.

Assignment 1:

Consider the same model as in pre-assignment but allow for panel structure in ε_{it} .

- (4 points) Carry out the between estimation and report the estimates, including standard errors and t-statistics, but also the coefficient of determination R^2 and your estimate of “ σ_α^2 ”.
- (4 points) Carry out the within estimation and report the estimates, including standard errors and t-statistics, but also the coefficient of determination R^2 and your estimate of “ σ_η^2 ”.
- (4 points) Carry out feasible GLS estimation and report the estimates, including standard errors and t-statistics, but also the coefficient of determination R^2 . Comment on your findings. How do the different estimates relate to each other ?
- (4 points) Carry out the Hausman test, what is your conclusion ?
- (4 points) Carry out the two-way fixed effects estimation and report the estimates, including standard errors and t-statistics, but also the coefficient of determination R^2 . What is your conclusion ?